

CSCI-338-Computer Science Theory
Fall 2025
HW 04

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Student Name:

This assignment is **due on November 04, 2025**. You will need to use Latex to generate a single pdf file and upload it under Assignment 04 on Canvas. There will be a penalty of not using Latex (to finish the assignment). This is not a group-assignment, so you must finish the assignment by yourself. **Figures can be hand drawn and inserted to the pdf.**

1 Problem 01

Using pumping lemma prove that the following languages are not context-free.

1. $L_1 = \{a^n b^j c^k \mid k = nj\}$
2. $L_2 = \{a^n b^j \mid n \geq (j-1)^3\}$

1. Assume L_1 is context-free. Let p be the pumping length. Choose the string

$$s = a^p b^p c^{p^2}.$$

This string is in L_1 because $k = p^2 = p \cdot p$.

By the pumping lemma, write $s = uvwxy$ with $|vwx| \leq p$ and $|vx| > 0$. Thus vwx lies entirely within one or two of the blocks of a 's, b 's, or c 's.

Let pumping increase or decrease counts of a, b, c by t_a, t_b, t_c respectively. After pumping with $i = 2$:

$$n' = p + t_a, \quad j' = p + t_b, \quad k' = p^2 + t_c.$$

For the pumped string to remain in L_1 we must have

$$k' = n'j' = (p + t_a)(p + t_b) = p^2 + p(t_a + t_b) + t_at_b.$$

But $|vwx| \leq p$ implies $t_c \leq p$, while the right-hand side requires

$$t_c = p(t_a + t_b) + t_at_b.$$

If $t_a + t_b \geq 1$ then $t_c \geq p$, and in fact $t_c > p$ unless $t_a = t_b = 0$, which is impossible because $|vx| > 0$. Thus $uv^2wx^2y \notin L_1$, violating the pumping lemma.

Therefore L_1 is not context-free.

2. Assume L_2 is context-free. Let p be the pumping length and choose:

$$s = a^{p^3}b^{p+1}.$$

This satisfies $p^3 \geq (p + 1 - 1)^3 = p^3$.

Write $s = uvwxy$ with the pumping lemma conditions. Consider cases:

Case A: v, x inside the a -block. Pumping down decreases n but keeps j unchanged, making

$$n' < (j' - 1)^3.$$

Contradiction.

Case B: v, x inside the b -block. Pumping up increases j so that

$$(j' - 1)^3 > p^3 = n'.$$

Contradiction.

Case C: v, x cross the block boundary. Pumping up increases j and increases n by at most p , but

$$(j' - 1)^3 \geq (p + 1)^3 > p^3 + p \geq n'.$$

Contradiction.

Thus L_2 is not context-free.

2 Problem 02

Let Π be the set of all infinite sequences over $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Show that Π is uncountable, using a proof by diagonalization.

Let $\mathbb{I}$ be the set of all infinite sequences over $\{0, 1, \dots, 9\}$. Prove $\mathbb{I}$ is uncountable using diagonalization.

Assume $\mathbb{I}$ is countable and list all sequences:

$$s^{(1)}, s^{(2)}, s^{(3)}, \dots, \quad s^{(k)} = (s_1^{(k)}, s_2^{(k)}, \dots).$$

Construct a new sequence t by:

$$t_i = \begin{cases} 0 & \text{if } s_i^{(i)} \neq 0, \\ 1 & \text{if } s_i^{(i)} = 0. \end{cases}$$

Then t differs from $s^{(i)}$ in the i th position, so t is not in the list. Contradiction. Thus $\mathbb{I}$ is uncountable.

3 Problem 03

Let $INFINITE_{PDA} = \{ \langle M \rangle \mid M \text{ is a PDA and } L(M) \text{ is an infinite language} \}$. Show that $INFINITE_{PDA}$ is decidable.

Let

$$INFINITE_PDA = \{ \langle M \rangle \mid M \text{ is a PDA and } L(M) \text{ is infinite} \}.$$

Show this language is decidable.

Convert the PDA M into an equivalent CFG G . It is decidable whether a CFG generates an infinite language:

1. Compute the productive variables of G (those that can derive terminal strings).
2. Build a graph whose nodes are productive variables, with $A \rightarrow B$ if $A \rightarrow \alpha B \beta$.
3. Check whether there is a cycle reachable from the start variable.

A reachable productive cycle implies infinitely many derivations; if none exists, $L(G)$ is finite. All steps are algorithmic, so $INFINITE_PDA$ is decidable.

4 Problem 04

Let $T = \{(i, j, k, l) \mid i, j, k, l \in \mathbb{N}\}$. Show that T is countable.

Let $T = \{(i, j, k, l) \mid i, j, k, l \in \mathbb{N}\}$. Show T is countable.

Enumerate by increasing the sum $s = i + j + k + l$. For $s = 0, 1, 2, \dots$, list all 4-tuples with sum s . Every tuple appears exactly once, giving a bijection with $\mathbb{N}$. Thus T is countable.

5 Problem 05

Show that if G is a CFG in Chomsky normal form, then for any string $w \in L(G)$ of length $n \geq 1$, exactly $2n - 1$ steps are required for any derivation of w .

To derive a string of length n :

- each of the n terminals requires one production $A \rightarrow a$,
- the parse tree is a full binary tree with n leaves, so it has $n - 1$ internal nodes,
- each internal node corresponds to a production $A \rightarrow BC$.

Thus total steps $= (n - 1) + n = 2n - 1$.

6 Problem 06

Let G be a CFG in Chomsky normal form that contains b variables. Show that if G generates some string with a derivation having at least 2^b steps, $L(G)$ is infinite.

Let G be a CFG in CNF with b variables. Show that if G has some derivation with at least $2b$ steps, then $L(G)$ is infinite.

A derivation with at least $2b$ steps produces a string of length

$$n \geq b + 1$$

by Problem 05.

Any derivation tree for such a string has a path from root to leaf containing at least $b + 1$ variables. By the pigeonhole principle, some variable A repeats on this path:

$$A \Rightarrow^+ uAv.$$

This is a pumpable pattern: repeating the cycle produces infinitely many distinct terminal strings. Thus $L(G)$ is infinite.